

Formal Logic (SM234614)

Third Meeting: Karnaugh Mapping

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Karnaugh mapping is a method for “simplifying” Boolean functions.

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Converting a truth table into a two-dimensional form

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Converting a truth table into a two-dimensional form

x_1	x_2	f
0	0	f_0
0	1	f_1
1	0	f_2
1	1	f_3

Karnaugh mapping is a method for “simplifying” Boolean functions.

Converting a truth table into a two-dimensional form

x_1	x_2	f
0	0	f_0
0	1	f_1
1	0	f_2
1	1	f_3

		x_2	
		0	1
x_1	0	f_0	f_1
	1	f_2	f_3

Karnaugh mapping is a method for “simplifying” Boolean functions.

Converting a truth table into a two-dimensional form

x_1	x_2	f
0	0	1
0	1	0
1	0	1
1	1	0

		x_2	
		0	1
x_1	0	1	0
	1	1	0

Karnaugh mapping is a method for “simplifying” Boolean functions.

Converting a truth table into a two-dimensional form

x_1	x_2	f
0	0	1
0	1	0
1	0	1
1	1	0

		x_2	
		0	1
x_1	0	1	0
	1	1	0

0: passive
1: active

0 active boxes: a contradiction

		x_2	
		0	1
x_1	0	0	0
	1	0	0

$f = \text{false}$

1 active box: a term involves two variables

		x_2	
		0	1
x_1	0	1	0
	1	0	0

$$f = \neg x_1 \neg x_2$$

2 active boxes (one row): a literal involves variable x_1

		x_2	
		0	1
x_1	0	1	1
	1	0	0

$$f = \neg x_1$$

2 active boxes (one column): a literal involves variable x_2

		x_2	
		0	1
x_1	0	1	0
	1	1	0

$$f = \neg x_2$$

4 active boxes: a tautology

		x_2	
		0	1
x_1	0	1	1
	1	1	1

$f = \text{true}$

		x_2	
		0	1
x_1	0	0	1
	1	1	0

$f =$

		x_2	
		0	1
x_1	0	0	1
	1	1	0

$$f = \neg x_1 x_2 + x_1 \neg x_2$$

		x_2	
		0	1
x_1	0	1	1
	1	1	0

$f =$

		x_2	
		0	1
x_1	0	1	1
	1	1	0

$$f = \neg x_1 + \neg x_2$$

Karnaugh Mapping with three variables

Karnaugh Mapping with three variables

		x_2x_3			
		00	01	11	10
x_1	0	f_0	f_1	f_3	f_2
	1	f_4	f_5	f_7	f_6

Karnaugh Mapping with three variables

		x_2x_3			
		00	01	11	10
x_1	0	f_0	f_1	f_3	f_2
	1	f_4	f_5	f_7	f_6

		x_3	
		0	1
x_1x_2	00	f_0	f_1
	01	f_2	f_3
	11	f_6	f_7
	10	f_4	f_5

0 active box: a contradiction

		x_2x_3			
		00	01	11	10
x_1	0	0	0	0	0
	1	0	0	0	0

$$f = \text{false}$$

1 active box: a term involves three variables

		x_2x_3			
		00	01	11	10
x_1	0	0	0	0	0
	1	0	0	0	1

$$f = x_1x_2\neg x_3$$

2 active boxes (one column): a term involves variables x_2 and x_3

		x_2x_3			
		00	01	11	10
x_1	0	0	0	0	1
	1	0	0	0	1

$$f = x_2 \neg x_3$$

2 active boxes (one row, consecutive locations): a term involves variables x_1 and x_2 or x_1 and x_3

		x_2x_3			
		00	01	11	10
x_1	0	1	1	0	0
	1	0	0	0	0

$$f = \neg x_1 \neg x_2$$

2 active boxes (one row, consecutive locations): a term involves variables x_1 and x_2 or x_1 and x_3

		x_2x_3			
		00	01	11	10
x_1	0	0	1	1	0
	1	0	0	0	0

$$f = \neg x_1 x_3$$

2 active boxes (one row, consecutive locations): a term involves variables x_1 and x_2 or x_1 and x_3

		x_2x_3			
		00	01	11	10
x_1	0	0	0	1	1
	1	0	0	0	0

$$f = \neg x_1 x_2$$

2 active boxes (one row, consecutive locations): a term involves the variables x_1 and x_2 or x_1 and x_3

		x_2x_3			
		00	01	11	10
x_1	0	1	0	0	1
	1	0	0	0	0

$$f = \neg x_1 \neg x_3$$

4 active boxes (one row): a literal involves the variable x_1

		x_2x_3			
		00	01	11	10
x_1	0	0	0	0	0
	1	1	1	1	1

$$f = x_1$$

4 active boxes (two rows, consecutive locations): a literal involves the variable x_2 or x_3

		x_2x_3			
		00	01	11	10
x_1	0	1	1	0	0
	1	1	1	0	0

$$f = \neg x_2$$

4 active boxes (two rows, consecutive locations): a literal involves the variable x_2 or x_3

		x_2x_3			
		00	01	11	10
x_1	0	0	1	1	0
	1	0	1	1	0

$$f = x_3$$

4 active boxes (two rows, consecutive locations): a literal involves the variable x_2 or x_3

		x_2x_3			
		00	01	11	10
x_1	0	0	0	1	1
	1	0	0	1	1

$$f = x_2$$

4 active boxes (two rows, consecutive locations): a literal involves the variable x_2 or x_3

		x_2x_3			
		00	01	11	10
x_1	0	1	0	0	1
	1	1	0	0	1

$$f = \neg x_3$$

16 active boxes: tautology

		x_2x_3			
		00	01	11	10
x_1	0	1	1	1	1
	1	1	1	1	1

$$f = \text{true}$$

		x_2x_3			
		00	01	11	10
x_1	0	0	1	1	1
	1	0	0	1	0

$$f = \dots$$

		x_2x_3			
		00	01	11	10
x_1	0	1	0	1	1
	1	1	1	0	1

$$f = \dots$$

Karnaugh Mapping with 4 variables

		x_3x_4			
		00	01	11	10
x_1x_2	00	f_0	f_1	f_3	f_2
	01	f_4	f_5	f_7	f_6
	11	f_8	f_9	f_{11}	f_{10}
	10	f_{12}	f_{13}	f_{15}	f_{14}

Karnaugh Mapping with 4 variables

	00	01	11	10
00	1	1	1	1
01	1	1	1	1
11	0	0	0	0
10	0	0	0	0

$f = \neg x_1$

Karnaugh Mapping with 4 variables

	00	01	11	10
00	1	1	1	1
01	1	1	1	1
11	0	0	0	0
10	0	0	0	0

$$f = \neg x_1$$

	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	1	1	1	1
10	1	1	1	1

$$f = x_1$$

Karnaugh Mapping with 4 variables

	00	01	11	10
00	1	1	1	1
01	1	1	1	1
11	0	0	0	0
10	0	0	0	0

$$f = \neg x_1$$

	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	1	1	1	1
10	1	1	1	1

$$f = x_1$$

	00	01	11	10
00	0	0	0	0
01	1	1	1	1
11	1	1	1	1
10	0	0	0	0

$$f = x_2$$

Karnaugh Mapping with 4 variables

	00	01	11	10
00	1	1	1	1
01	1	1	1	1
11	0	0	0	0
10	0	0	0	0

$$f = \neg x_1$$

	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	1	1	1	1
10	1	1	1	1

$$f = x_1$$

	00	01	11	10
00	0	0	0	0
01	1	1	1	1
11	1	1	1	1
10	0	0	0	0

$$f = x_2$$

	00	01	11	10
00	1	1	1	1
01	0	0	0	0
11	0	0	0	0
10	1	1	1	1

$$f = \neg x_2$$

Karnaugh Mapping with 4 variables

	00	01	11	10
00	1	1	1	1
01	1	1	1	1
11	0	0	0	0
10	0	0	0	0

$$f = \neg x_1$$

	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	1	1	1	1
10	1	1	1	1

$$f = x_1$$

	00	01	11	10
00	0	0	0	0
01	1	1	1	1
11	1	1	1	1
10	0	0	0	0

$$f = x_2$$

	00	01	11	10
00	1	1	1	1
01	0	0	0	0
11	0	0	0	0
10	1	1	1	1

$$f = \neg x_2$$

	00	01	11	10
00	1	1	0	0
01	1	1	0	0
11	1	1	0	0
10	1	1	0	0

$$f = \neg x_3$$

Karnaugh Mapping with 4 variables

	00	01	11	10
00	1	1	1	1
01	1	1	1	1
11	0	0	0	0
10	0	0	0	0

$$f = \neg x_1$$

	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	1	1	1	1
10	1	1	1	1

$$f = x_1$$

	00	01	11	10
00	0	0	0	0
01	1	1	1	1
11	1	1	1	1
10	0	0	0	0

$$f = x_2$$

	00	01	11	10
00	1	1	1	1
01	0	0	0	0
11	0	0	0	0
10	1	1	1	1

$$f = \neg x_2$$

	00	01	11	10
00	1	1	0	0
01	1	1	0	0
11	1	1	0	0
10	1	1	0	0

$$f = \neg x_3$$

	00	01	11	10
00	0	0	1	1
01	0	0	1	1
11	0	0	1	1
10	0	0	1	1

$$f = x_3$$

Karnaugh Mapping with 4 variables

	00	01	11	10
00	1	1	1	1
01	1	1	1	1
11	0	0	0	0
10	0	0	0	0

$$f = \neg x_1$$

	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	1	1	1	1
10	1	1	1	1

$$f = x_1$$

	00	01	11	10
00	0	0	0	0
01	1	1	1	1
11	1	1	1	1
10	0	0	0	0

$$f = x_2$$

	00	01	11	10
00	1	1	1	1
01	0	0	0	0
11	0	0	0	0
10	1	1	1	1

$$f = \neg x_2$$

	00	01	11	10
00	1	1	0	0
01	1	1	0	0
11	1	1	0	0
10	1	1	0	0

$$f = \neg x_3$$

	00	01	11	10
00	0	0	1	1
01	0	0	1	1
11	0	0	1	1
10	0	0	1	1

$$f = x_3$$

	00	01	11	10
00	0	1	1	0
01	0	1	1	0
11	0	1	1	0
10	0	1	1	0

$$f = x_4$$

Karnaugh Mapping with 4 variables

	00	01	11	10
00	1	1	1	1
01	1	1	1	1
11	0	0	0	0
10	0	0	0	0

$$f = \neg x_1$$

	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	1	1	1	1
10	1	1	1	1

$$f = x_1$$

	00	01	11	10
00	0	0	0	0
01	1	1	1	1
11	1	1	1	1
10	0	0	0	0

$$f = x_2$$

	00	01	11	10
00	1	1	1	1
01	0	0	0	0
11	0	0	0	0
10	1	1	1	1

$$f = \neg x_2$$

	00	01	11	10
00	1	1	0	0
01	1	1	0	0
11	1	1	0	0
10	1	1	0	0

$$f = \neg x_3$$

	00	01	11	10
00	0	0	1	1
01	0	0	1	1
11	0	0	1	1
10	0	0	1	1

$$f = x_3$$

	00	01	11	10
00	0	1	1	0
01	0	1	1	0
11	0	1	1	0
10	0	1	1	0

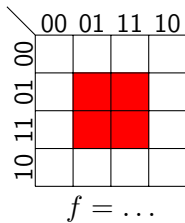
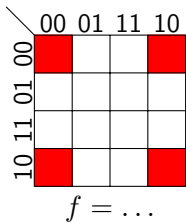
$$f = x_4$$

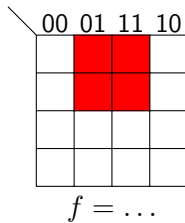
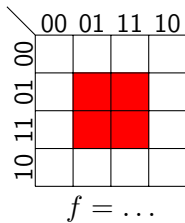
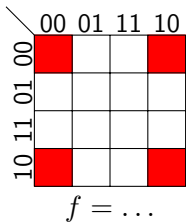
	00	01	11	10
00	1	0	0	1
01	1	0	0	1
11	1	0	0	1
10	1	0	0	1

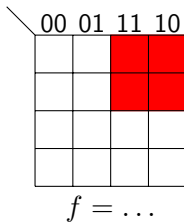
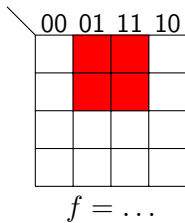
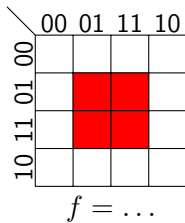
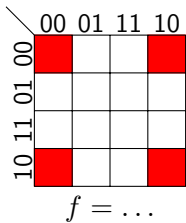
$$f = \neg x_4$$

	00	01	11	10
00	1	0	0	1
01	0	0	0	0
11	0	0	0	0
10	1	0	0	1

$f = \dots$







	00	01	11	10
00				
01				
11				
10				

$f = \dots$

	00	01	11	10
00				
01				
11				
10				

$f = \dots$

	00	01	11	10
00				
01				
11				
10				

$f = \dots$

	00	01	11	10
00				
01				
11				
10				

$f = \dots$

	00	01	11	10
00	0	1	1	0
01	0	1	0	1
11	0	0	1	1
10	0	1	1	0

$f = \dots$

	00	01	11	10
00				
01				
11				
10				

$f = \dots$

	00	01	11	10
00				
01				
11				
10				

$f = \dots$

	00	01	11	10
00				
01				
11				
10				

$f = \dots$

	00	01	11	10
00				
01				
11				
10				

$f = \dots$

	00	01	11	10
00	0	1	1	0
01	0	1	0	1
11	0	0	1	1
10	0	1	1	0

$f = \dots$

	00	01	11	10
00	1	0	0	1
01	0	0	0	0
11	1	1	0	1
10	0	1	1	1

$f = \dots$

	00	01	11	10
00				
01				
11				
10				

$f = \dots$

	00	01	11	10
00				
01				
11				
10				

$f = \dots$

	00	01	11	10
00				
01				
11				
10				

$f = \dots$

	00	01	11	10
00				
01				
11				
10				

$f = \dots$

	00	01	11	10
00	0	1	1	0
01	0	1	0	1
11	0	0	1	1
10	0	1	1	0

$f = \dots$

	00	01	11	10
00	1	0	0	1
01	0	0	0	0
11	1	1	0	1
10	0	1	1	1

$f = \dots$

	00	01	11	10
00	1	1	0	0
01	1	0	0	0
11	0	1	1	1
10	1	1	0	1

$f = \dots$

	00	01	11	10
00				
01				
11				
10				

$f = \dots$

	00	01	11	10
00				
01				
11				
10				

$f = \dots$

	00	01	11	10
00				
01				
11				
10				

$f = \dots$

	00	01	11	10
00				
01				
11				
10				

$f = \dots$

	00	01	11	10
00	0	1	1	0
01	0	1	0	1
11	0	0	1	1
10	0	1	1	0

$f = \dots$

	00	01	11	10
00	1	0	0	1
01	0	0	0	0
11	1	1	0	1
10	0	1	1	1

$f = \dots$

	00	01	11	10
00	1	1	0	0
01	1	0	0	0
11	0	1	1	1
10	1	1	0	1

$f = \dots$

	00	01	11	10
00	0	0	1	1
01	1	0	0	1
11	1	0	1	0
10	1	0	1	1

$f = \dots$